

Re: Grant - power calculations

Summary: Using preliminary RNA-seq data from four donors to estimate the variability of IRAK3 expression, we estimated the effect size detectable with 80% power for the proposed sample size of 10 normal and 10 severe asthma donors. Assuming comparable variability in the future cohort, the study would have 80% power to detect approximately 1.2–1.6-fold differences in IRAK3 expression between groups, depending on treatment and cell type.

These estimates are based on preliminary data from only four donors and therefore should be considered approximate; they assume that between-donor variability in the planned cohort is comparable to that observed in the preliminary dataset.

Example calculation

Pull IRAK3 data for HBEs at NS:

```
## # A tibble: 5 × 9
##   Gene rep    celltype treatment  time   tpm log2tpm  fold log2fold
##   <chr> <chr>   <fct>    <fct>    <dbl> <dbl>   <dbl> <dbl>   <dbl>
## 1 IRAK3 donor67 HBE      NS        6 0.475 -0.797    1      0
## 2 IRAK3 donor75 HBE      NS        6 0.415 -0.959    1      0
## 3 IRAK3 donor74 HBE      NS        6 0.657 -0.401    1      0
## 4 IRAK3 donor86 HBE      NS        6 0.965  0.0909    1      0
## 5 IRAK3 donor90 HBE      NS        6 0.485 -0.774    1      0
```

Calculate standard deviation of log2tpm values:

```
## [1] 0.4212214
```

Given a two-sample design, N=10, alpha=0.05, and standard deviation from above, what is the power to detect a 2-fold (delta=1 in log2 space) change between normal and asthmatic?

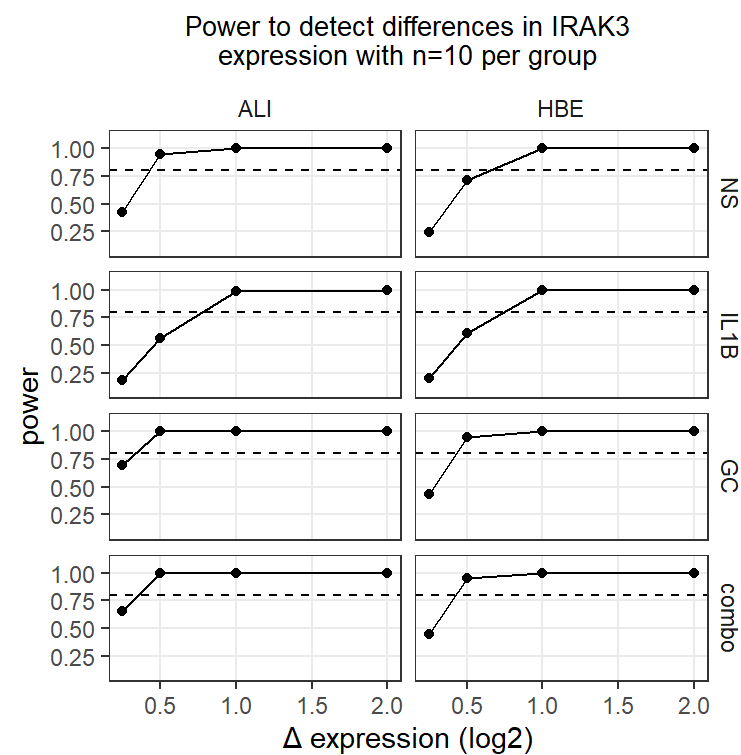
```
##
##   Two-sample t test power calculation
##
##           n = 10
##         delta = 1
##        sd = 0.4212214
##   sig.level = 0.05
##        power = 0.9988508
## alternative = two.sided
##
## NOTE: n is number in *each* group
```

So, 99.9% probability of correctly detecting a doubling of IRAK3 expression between groups. This does not mean that the study has 99.9% power overall.

Preliminary power analysis for between-group differences in IRAK3 expression

We can scale out the above calculation to IRAK3 in all primary cell types, across a range of gene expression changes:

Based on the variability observed in our preliminary donors, 10 donors per group provides $\geq 80\%$ power to detect differences in approximately the 1.4–2-fold range (depending on cell type and treatment). Dotted line at power=0.8 for visualization.



Explicitly calculating the effect size detectable at 80% power, the proposed sample size of 10 normal and 10 severe asthma donors provides 80% power to detect approximately 1.2–1.6-fold differences in IRAK3 expression, depending on treatment and cell type.

treatment	celltype	sd_estimate	power	log2delta	fold_change
NS	ALI	0.2993301	0.8	0.3965961	1.316398
NS	HBE	0.4212214	0.8	0.5580955	1.472324
IL1B	ALI	0.5017958	0.8	0.6648521	1.585406
IL1B	HBE	0.4732822	0.8	0.6270733	1.544429
GC	ALI	0.2138748	0.8	0.2833725	1.217037
GC	HBE	0.2936747	0.8	0.3891030	1.309579
combo	ALI	0.2245432	0.8	0.2975076	1.229019
combo	HBE	0.2889613	0.8	0.3828580	1.303922